Hemanjan Reddy Pundla

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Technical Skills

Languages: Python, Java, SQL, JavaScript/TypeScript
Backend & Systems: REST APIs, FastAPI, Flask, microservices, event-driven architecture, distributed systems, PostgreSQL
Cloud & Infrastructure: AWS (EKS, Lambda, Step Functions, S3, OpenSearch), Kubernetes, Docker, Terraform, Helm
CI/CD & Observability: Jenkins, GitHub Actions, ArgoCD, Prometheus, Grafana, pytest, JUnit
Applied AI: Amazon Bedrock, RAG, vector search, multi-agent orchestration, LLM evaluation

Experience

Capital One Apr 2026 – Present
Senior Software Engineer *McLean, VA (Hybrid)*

- Build Python/FastAPI backend services for **Prometheus Financial Core (PFC)**, a shared GenAI platform exposing LLM and retrieval capabilities via REST APIs to 4 business areas (Card, Bank, Core, Internal), serving ∼50K requests/day with retrieval p99 under 800 ms
- Designed a multi-agent orchestration service on Amazon Bedrock that automates loan-document review and data extraction, cutting analyst handling time from ∼45 to under 8 minutes per file with a full audit trail
- Built a RAG retrieval layer on OpenSearch over 25K+ policy and product documents, returning citation-backed answers and raising grounded-answer accuracy from 72% to 89% on an internal eval set
- Deploy services to EKS with Helm and ArgoCD on Terraform-managed infrastructure; built Jenkins pipelines that gate every deploy on unit and integration tests, cutting merge-to-production time from ∼1 day to under 1 hour
- Own service health with Prometheus/Grafana alerting and A/B evaluation of model changes; led root-cause analysis on 5 production incidents, sustaining 99.9% availability against SLO

Neighborly Nov 2024 – Mar 2026
Software Development Engineer *Irving, TX*

- Built backend services and REST APIs for a CSR-facing lead-management platform that surfaces unattended leads in real time across 12 home-service brands, tracking ∼40K leads/month
- Designed the unattended-lead workflow (queues, state transitions, SLA-based alerts) giving agents one view of open, stale, and abandoned leads, reducing median lead close time by 18%
- Built scheduled sync jobs between the CRM and CSR console with idempotent retries, cutting duplicate customer outreach by ∼30% and removing most manual lookups
- Defined lead SLA rules with call-center operations and product; instrumented supervisor dashboards that surface backlogs within 5 minutes
- Owned delivery end to end (design docs, code review, tests, on-call rotation) and improved p95 API latency from 850 ms to 320 ms through query tuning and caching

University of North Texas Oct 2023 – Jul 2024
AI/ML Research Assistant *Denton, TX*

- Built and evaluated a BERT-based text-classification pipeline on a 50K-document dataset, improving macro-F1 from 0.78 to 0.86 over a TF-IDF + logistic regression baseline

BrandSafway Apr 2021 – Mar 2023
Software Development Engineer *India*

- Built Python REST APIs (Flask) for equipment-rental and workforce-scheduling systems used across 150+ branches by ∼5K internal users
- Rewrote slow SQL queries and added indexes on the reporting data-access layer, cutting report latency from ∼12s to under 2s for analytics consumers
- Owned the equipment check-in/check-out module from design to release, reducing manual inventory reconciliation by ∼40%
- Raised test coverage from 45% to 80% with pytest suites in CI, reducing production regressions by ∼35%

Projects

AuraWorx – Enterprise AI Platform | *LibreChat, Python, FastAPI, Docker*

- Extended LibreChat into a multi-tenant enterprise assistant with a document-OCR pipeline (Tesseract) processing 500+ pages/hour and an admin dashboard for tenant, user, and usage management
- Integrated live web search (Tavily API) so answers cite current sources, improving answer accuracy on time-sensitive questions by ∼35% in internal testing, with per-tenant isolation of documents and chat history

Lifestyle Coach | *Python, FastAPI, LLM*

- Built an AI coach that gives personalized fitness, nutrition, and habit guidance from each user's goals and profile, with median response time under 2s
- Designed the conversational pipeline and a prompt/retrieval layer that stores user goals and history, so advice adapts across sessions; cut token usage per conversation by ∼40% through context summarization

Education

University of North Texas | M.S. in Artificial Intelligence — Aug 2023 – May 2025

JNTU Anantapur | B.Tech. in Electronics and Communications Engineering — 2017 – 2021